PARSHWANATH CHARITABLE TRUST'S
A.P. SHAH INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Data Science

UniAccess Network Analyzer

A Mobile Application for Network Access Management

Kunal Singh Thakur (24107027)
Parth Mane (24107127)
Pratham Shetty (24107088)
Tanmay Monde (24107034)

Project Guide
Ms. Rajashri Chaudhari

Contents

- Introduction
- Objectives
- Scope
- Features / Functionality
- Project Outcomes
- Technology Stack
- Block Diagram (if applicable)

1. Introduction

Problem Identified :

- Managing and controlling network usage in shared spaces like universities is difficult without a dedicated tool.

- Administrators have no simple mobile interface to monitor connected devices, detect misuse, or apply access control in real time.

Solution Proposed :

- UniAccess is a mobile first system where administrators can view and manage network usage from a single Flutter app.

- Traffic from connected devices is observed using Wireshark to understand usage patterns such as normal and unusually high activity. The app acts as a control interface allowing admins to view device activity, apply access rules, and block or restrict devices when required.

2. Objectives

1. To provide a simple mobile solution for managing network access in shared environments such as universities.
2. To observe and compare normal and high network usage among connected devices.
3. To help administrators identify devices that misuse network resources.
4. To enable quick access control decisions through an easy-to-use mobile interface.

3. Scope

1. Can be applied in university campus networks, hostels, computer labs, and other shared Wi-Fi environments.
2. Can be useful to network administrators, IT staff, and department heads who need quick visibility into network activity.
3. Can be extended to support multiple admin roles and larger network environments with more connected devices.
4. Can be used to enforce data usage policies by identifying and restricting high-consumption devices.
5. Can be integrated with existing network infrastructure.

4. Features / Functionality

1. Mobile Dashboard :

Shows all connected devices with their usage status IP address, ping, packets sent, and files uploaded updated in real time.

2. Device Classification :

Automatically categorizes devices as normal or high usage based on packet count and activity observed through Wireshark & Tshark packet capture.

3. Access Control :

Admin can instantly restrict or block any connected device from the mobile app. Blocked devices are disconnected and prevented from reconnecting.

4. Activity & Log History :

Records all network events connections, disconnections, file uploads with timestamps and IP addresses. Logs stored in PostgreSQL Storage.

5. Attacker Autoblock :

At a certain threshold UniAccess will detect an unusual amount of network traffic leading to the auto block feature to block that port and ultimately the access to the network via that port**.**

5. Outcome of Project

1. Admin can register, login, and monitor the entire network from a secure mobile dashboard.

2. Admin can view all connected client devices with real-time statistics including IP address, ping, and packet count.

3. Admin can instantly block or unblock any device, blocked devices are force disconnected and prevented from reconnecting.

4. The system automatically detects and highlights high-traffic or flood-generating devices, alerting the admin visually.

5. Admin can view live Wireshark style packet logs showing all network activity in real time.

6. Admin can download activity history logs for any date for record keeping and analysis.

Sustainable Development Goals (SDG) mapped

SDG 9: Industry, Innovation, and Infrastructure.

- Promotes development of secure and reliable digital infrastructure.
- Enhances network efficiency and stability in institutions.
- Supports innovation in real-time network monitoring systems.

SDG 4: Quality Education.

- Ensures uninterrupted internet access for students.
- Improves learning experience by preventing network slowdowns.
- Supports digital learning environments in universities.

SDG 16: Project Contribution.

- Detects and prevents network misuse (flood attacks)Maintains fair usage of internet resources.
- Provides real-time monitoring and quick response system.
- Enhances overall digital infrastructure security.

6. Technology Stack

Component	Technology / Details
Frontend (GUI)	Flutter (Dart) — Cross-platform mobile application
Backend Server	Python FastAPI — REST API & WebSocket server
Database	PostgreSQL— Flexible
Real-time Comms	WebSocket — Data push to mobile app
Packet Capture	Wireshark(Tshark) — Network packet monitoring and analysis
Authentication	Secure login and session management (Network Engineer .edu access)

7. Block Diagram

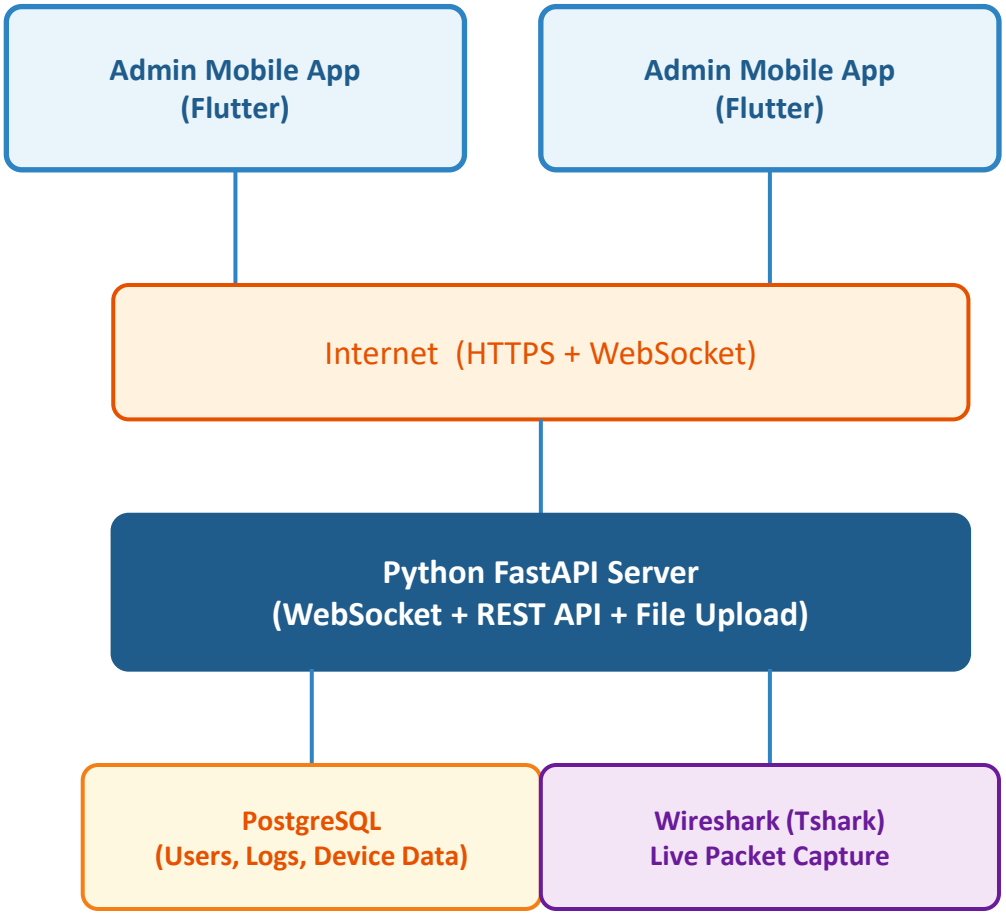
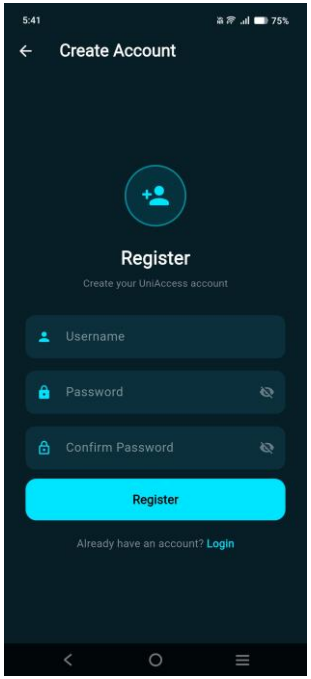
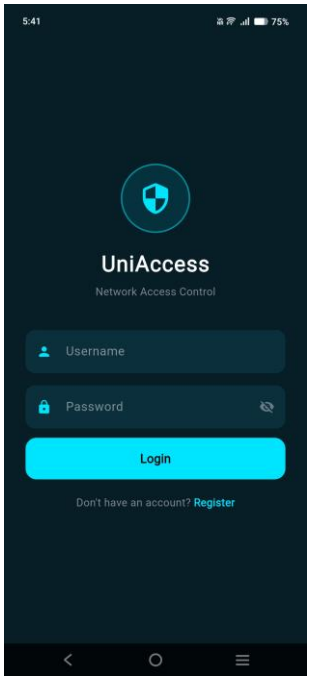
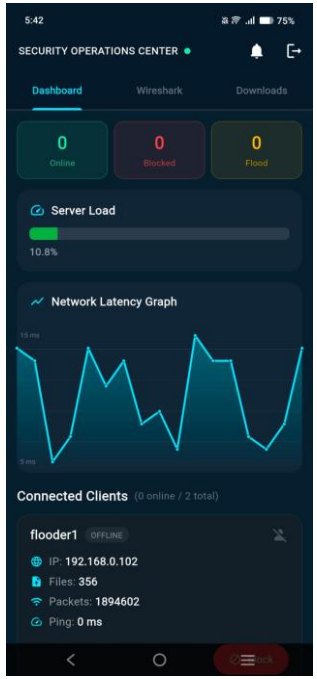


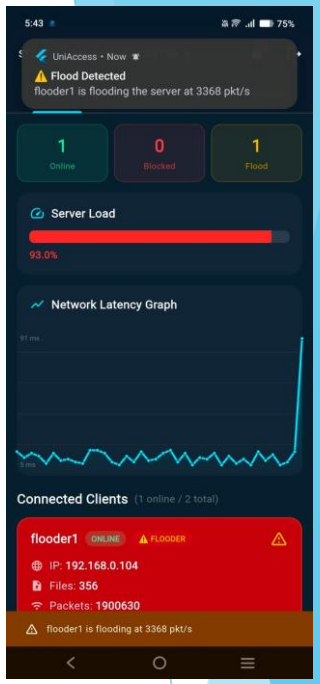
Figure: UniAccess System Architecture

Output:

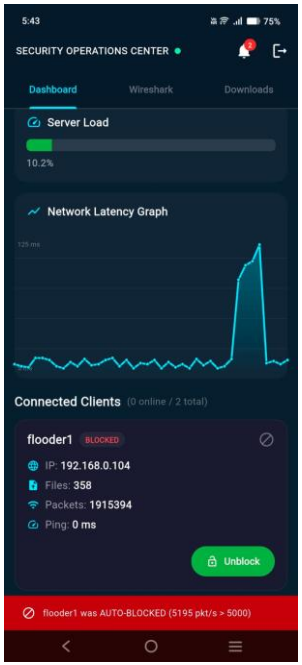

Create Account
Register
Create your UniAccess account
Username
Password
Confirm Password
Register
Already have an account? Login

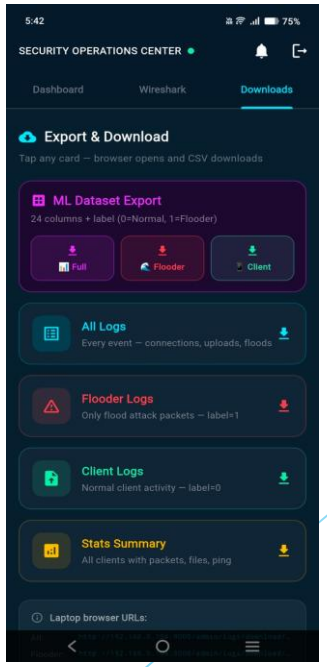

UniAccess
Network Access Control
Username
Password
Login
Don't have an account? Register


SECURITY OPERATIONS CENTER
Dashboard
Wireshark
Downloads
0
Online
0
Blocked
0
Flood
Server Load
10.8%
Network Latency Graph
Connected Clients (0 online / 2 total)
flooder1 OFFLINE
IP: 192.168.0.102
Files: 356
Packets: 1894602
Ping: 0 ms


UniAccess • Now
Flood Detected
flooder1 is flooding the server at 3368 pkt/s
1
Online
0
Blocked
1
Flood
Server Load
93.0%
Network Latency Graph
Connected Clients (1 online / 2 total)
flooder1 ONLINE FLOODER
IP: 192.168.0.104
Files: 356
Packets: 1900630
flooder1 is flooding at 3368 pkt/s

SECURITY OPERATIONS CENTER
Dashboard
Wireshark
Downloads
Wireshark Packet Analyzer
100 pkts
CSV


SECURITY OPERATIONS CENTER
Dashboard
Wireshark
Downloads
Server Load
10.2%
Network Latency Graph
Connected Clients (0 online / 2 total)
flooder1 BLOCKED
IP: 192.168.0.104
Files: 358
Packets: 1915394
Ping: 0 ms
Unblock
flooder1 was AUTO-BLOCKED (5195 pkt/s > 5000)


SECURITY OPERATIONS CENTER
Dashboard
Wireshark
Downloads
Export & Download
Tap any card — browser opens and CSV downloads
ML Dataset Export
24 columns + label (0=Normal, 1=Flooder)
Full
Flooder
Client
All Logs
Every event — connections, uploads, floods
Flooder Logs
Only flood attack packets — label=1
Client Logs
Normal client activity — label=0
Stats Summary
All clients with packets, files, ping
Laptop browser URLs:

Thank You.